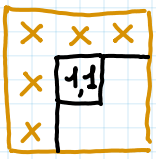
PROBLEMA nel BORDO



Possibile soluzione:
- riempire con ZERI → ZERO-PADDING il bordo
- ridurre l'immagine di output

$$H \times W \xrightarrow{3\times 3} (H-2) \times (W-2)$$

- estendere le intensità dei pixel sul bordo

Si calcolano
$$L_x = L * M_x$$
$$L_y = L * M_y$$

$$\nabla L = \left(\frac{\partial L}{\partial x}, \frac{\partial L}{\partial y}\right) \cong (L_x, L_y)$$ COME OTTENERE L'OUTPUT?

$$\|\nabla L\| = \sqrt{\left(\frac{\partial L}{\partial x}\right)^2 + \left(\frac{\partial L}{\partial y}\right)^2}$$ è una possibilità

SOBEL: $$\boxed{G_L = |L_x| + |L_y|}$$ FILTRO GRADIENTE

↳ per "raffinare" la ricerca degli EDGE:
- CANNY EDGE DETECTOR ...

FILTRO LAPLACIANO

Def: LAPLACIANO di $f: \mathbb{R}^2 \to \mathbb{R}$ t.c. $f \in \mathcal{C}^2(\mathbb{R}^2)$ è dato da:

$$\nabla^2 f := \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

(è diverso da $(\nabla f)^2$!!)

APROX di f al II ORDINE con LAPLACIANO

$\underline{x}_0 = (x, y)$ PIXEL centrale e $\underline{x} = (x+h, y+k)$ con $h, k \in \{-1, 0, 1\}$

$$f(x+h, y+k) \overset{T}{\cong} \underbrace{f(x,y)}_{\alpha=(0,0)} + \underbrace{\frac{\partial f}{\partial x}(x,y) \cdot h}_{\alpha=(1,0)} + \underbrace{\frac{\partial f}{\partial y}(x,y) \cdot k}_{\alpha=(0,1)}$$

$$\alpha=(2,0) \quad + \frac{\partial^2 f}{\partial x^2}(x,y) \cdot \frac{h^2}{2} + \frac{\partial^2 f}{\partial y^2}(x,y) \cdot \frac{k^2}{2} + \quad \alpha=(0,2)$$

$$\alpha=(1,1)=(1,1) \quad + \cancel{2} \cdot \frac{\partial^2 f}{\partial x \partial y}(x,y) \cdot \frac{h \cdot k}{\cancel{2}}$$

① $h=1,\ k=0$ $\quad f(x \pm 1, y) \cong f(x,y) \pm \frac{\partial f}{\partial x}(x,y) + \frac{\partial^2 f}{\partial x^2}(x,y) \cdot \frac{1}{2}$

② $h=-1,\ k=0$

① + ②: $\quad f(x+1,y) + f(x-1,y) \cong 2f(x,y) + \underbrace{\frac{\partial^2 f}{\partial x^2}(x,y)}$

③ $h=0,\ k=-1$ $\quad f(x, y \pm 1) \cong f(x,y) \pm \frac{\partial f}{\partial y}(x,y) + \frac{\partial^2 f}{\partial y^2}(x,y) \cdot \frac{1}{2}$

④ $h=0,\ k=1$

③ + ④: $\quad f(x,y-1) + f(x,y+1) \cong 2f(x,y) + \underbrace{\frac{\partial^2 f}{\partial y^2}(x,y)}$

$$\frac{\partial^2 f}{\partial x^2}(x,y) \cong f(x-1,y) - 2f(x,y) + f(x+1,y)$$

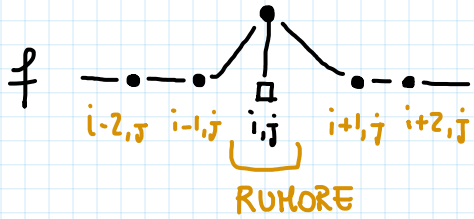
$$\frac{\partial^2 f}{\partial y^2}(x,y) \cong f(x,y-1) - 2f(x,y) + f(x,y+1)$$

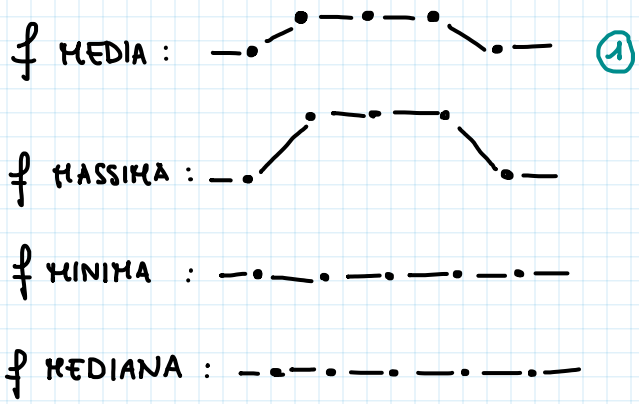
$$M_{xx} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & -2 & 0 \\ 0 & 1 & 0 \end{pmatrix} \qquad M_{yy} = \begin{pmatrix} 0 & 0 & 0 \\ 1 & -2 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} \qquad M_{\nabla^2} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$\nabla^2 L = L * (\underbrace{M_{xx} + M_{yy}}_{M_{\nabla^2}})$$

FILTRI di SMOOTHING ⟶ riduzione rumore





(1)

(1) Dimensione m dispari $m \in \mathbb{N}$, def. FILTRO di MEDIA

$$M_m = \begin{pmatrix} \frac{1}{m^2} & \frac{1}{m^2} & \cdots \\ & \vdots & \\ & & \frac{1}{m^2} \end{pmatrix}$$

per es $M_3 = \begin{pmatrix} 1/9 & 1/9 & 1/9 \\ 1/9 & 1/9 & 1/9 \\ 1/9 & 1/9 & 1/9 \end{pmatrix}$

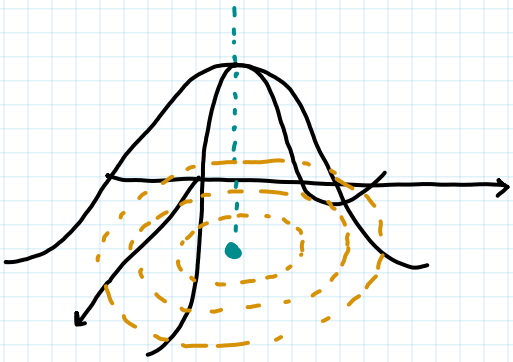
⟶ FILTRO LINEARE

FILTRO GAUSSIANO :

$$f(x,y) = \frac{1}{2\pi\sigma^2} e^{-\frac{(x^2+y^2)}{2\sigma^2}} \qquad \mu = 0 \;,\; \sigma = 1$$

$$\boxed{M_{ij} = \frac{1}{2\pi} e^{-\frac{(i^2+j^2)}{2}}}$$

$M \in \mathbb{R}^{m \times m}$ con m dispari



⟶ LINEARE

$$M = \begin{pmatrix} i-1,j-1 & i,j-1 & i+1,j-1 \\ i-1,j & i,j & i+1,j \\ i-1,j+1 & i,j+1 & i+1,j+1 \end{pmatrix}$$